



Flink state evolution

Flink serialization mechanisms,
and how to evolve the state's structure

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Speakers



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Scraping



Account
Takeover



Credential
Stuffing



Payment



Fake Account
Creation



Scalper Bots



Denial of
Service



Vulnerability
scanning



Ad/Click
Fraud



Skewed
Analytics

Flink at DataDome

300K

Events/second

7

Flink jobs

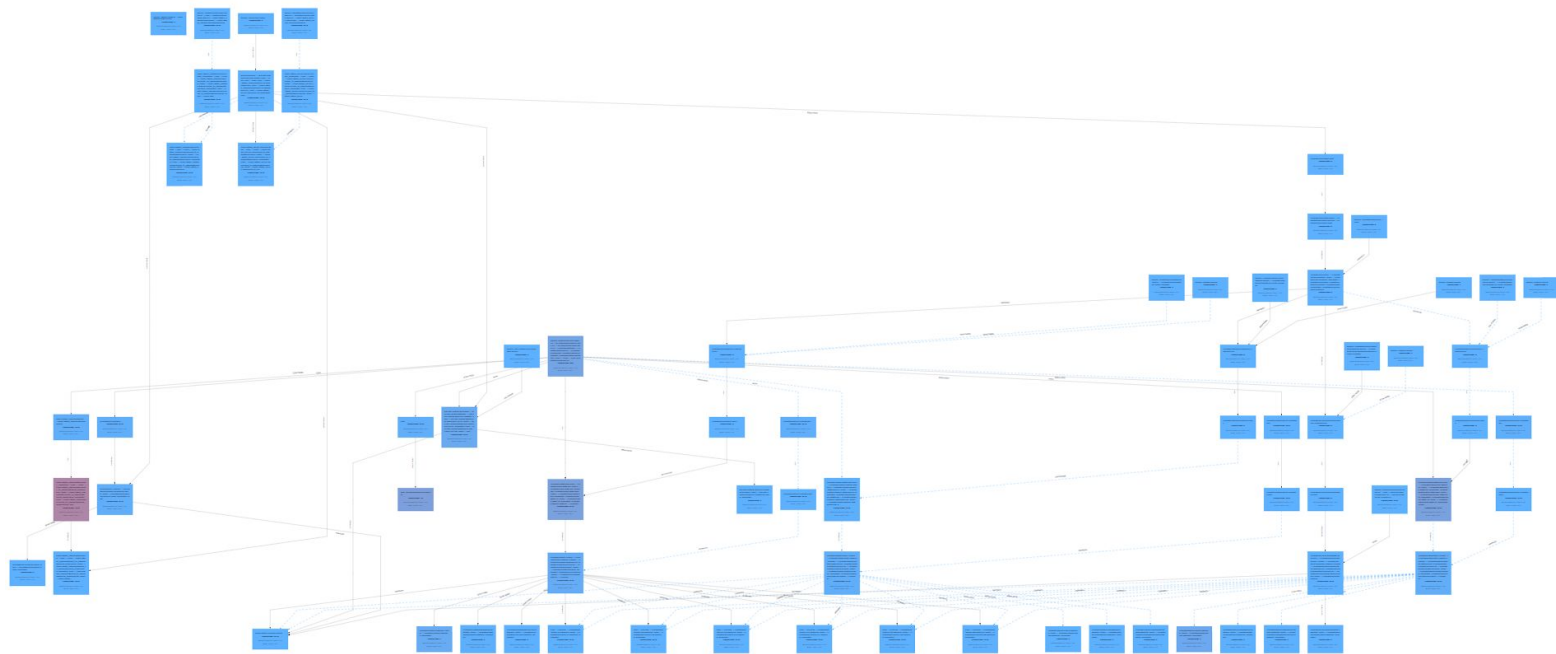
~330

Task-managers

~30

Deployments/month

Flink at DataDome



The problem

Upgrading...

2026-09-14 14:10:43

✓ java.lang.Exception: Except

at org.apache.flink.str

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Caused by: java.lang.RuntimeException: Error while getting state

at org.apache.flink.runtime.state.DefaultKeyedStateStore.getState(DefaultKeyedStateStore.java:62)

at org.apache.flink.streaming.api.operators.StreamingRuntimeContext.getState(StreamingRuntimeContext.java:203)

at co.datadome.sherlock.newVerdicts.rules.IndexBuildFunction.buildPostponedCountStates\$lzycompute(IndexBuildFunction.scala:74)

at co.datadome.sherlock.newVerdicts.rules.IndexBuildFunction.buildPostponedCountState(IndexBuildFunction.scala:74)

at co.datadome.sherlock.newVerdicts.rules.IndexBuildFunction.getBuildPostponedCount(IndexBuildFunction.scala:76)

at co.datadome.sherlock.newVerdicts.rules.IndexBuildFunction.onTimer(IndexBuildFunction.scala:91)

at org.apache.flink.streaming.api.operators.KeyedProcessOperator.invokeUserFunction(KeyedProcessOperator.java:294)

at org.apache.flink.streaming.api.operators.KeyedProcessOperator.onProcessingTime(KeyedProcessOperator.java:294)

at org.apache.flink.streaming.runtime.tasks.StreamTask.invokeProcessingTimeCall(StreamTask.java:712)

at org.apache.flink.streaming.runtime.tasks.StreamTask.restore(StreamTask.java:712)

at org.apache.flink.streaming.runtime.taskmanager.Task.runWithSystemExitMonitoring(Task.java:958)

at org.apache.flink.streaming.runtime.taskmanager.Task.restoreAndInvoke(Task.java:927)

at org.apache.flink.streaming.runtime.taskmanager.Task.doRun(Task.java:751)

at org.apache.flink.streaming.runtime.taskmanager.Task.run(Task.java:566)

at java.base/java.lang.Thread.run(Thread.java:1583)

Caused by: org.apache.flink.util.StateMigrationException: For heap backends, the

flink.api.common.typeutils.base.IntSerializer@3df6911a).

at org.apache.flink.runtime.state.heap.HeapKeyedStateBackend.tryRegister(HeapKeyedStateBackend.java:1583)

at org.apache.flink.runtime.state.heap.HeapKeyedStateBackend.createOrRestore(HeapKeyedStateBackend.java:1583)

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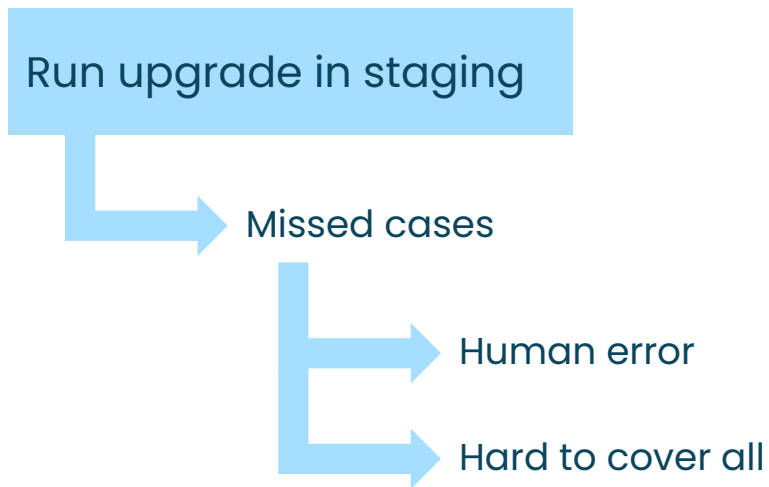
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« Just test it ! »



Early solution: State signature

State evolution $\Rightarrow$ Operator UID reset



[Flink Forward 2020 Handling frequent changes in state structure](#)



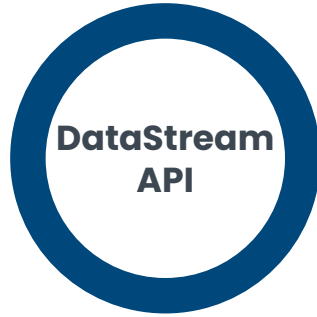
Not a solution: Flink's State Processor API

Upgrade the savepoint before redeploying

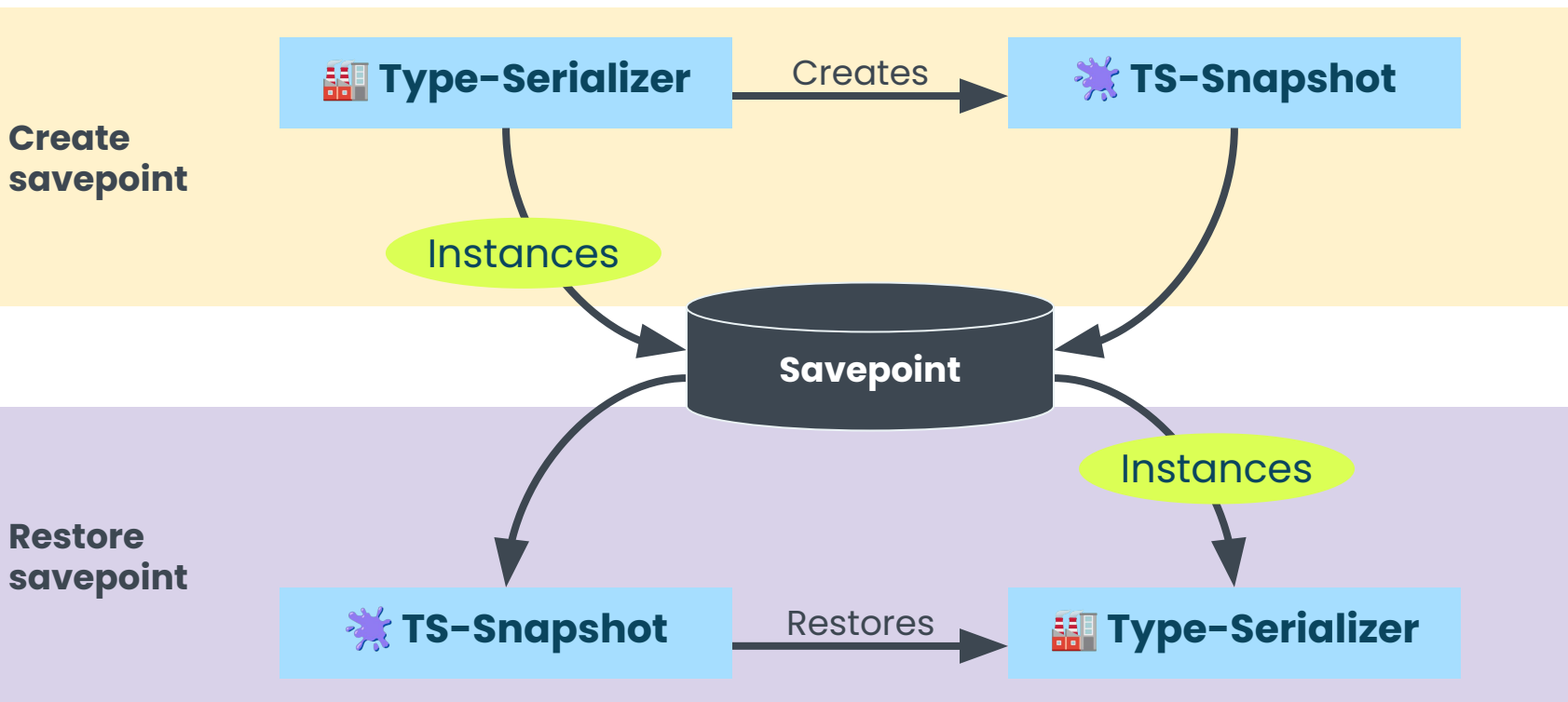


How serialization works

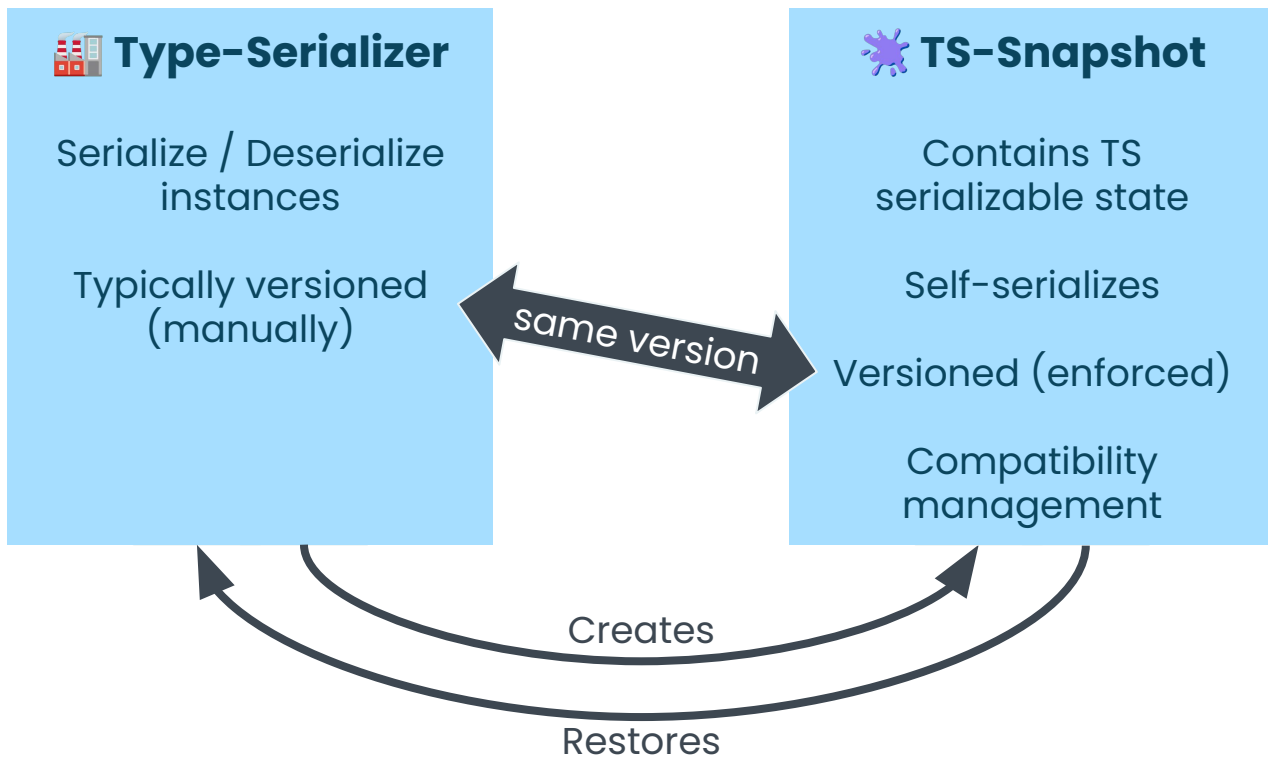
Caveats



Type-Serializer and Type-Serializer-Snapshot



Type-Serializer and Type-Serializer-Snapshot



When creating a savepoint

Register TS

```
DESC = new ValueStateDescriptor<Stats>(  
    "stats",  
    new StatsSerializer()  
);
```

My job

Flink

Serialize TSS

```
snap = ser.snapshotConfiguration();
```

```
out.writeUTF(snap.getClass().getName());  
out.writeInt(snap.getCurrentVersion());  
snap.writeSnapshot(out);
```

Serialize state

```
ser.serialize(...);
```

Passes the version number

No need to serialize the version manually

Uses the latest version by default

When restarting from a savepoint

Requires a no-arg constructor !

Restore TSS then TS

```
className = in.readUTF();  
clazz = Class.forName(className, ...);  
oldSnap = clazz.newInstance();  
version = in.readInt();  
oldSnap.readSnapshot(version, in, ...);
```

```
oldSer = oldSnap.restoreSerializer();
```

Passes the version number

Register TS and TSS

```
DESC = new ValueStateDescriptor<Stats>(  
    "stats",  
    new StatsSerializer() // newSer  
);
```

```
newSnap = newSer.snapshotConfiguration();
```

May be different classes

Check compatibility

```
newSnap.resolveSchemaCompatibility(oldSnap);
```

Deserialize state

```
ser.deserialize(...);
```

We'll get back to that one

Heap vs RocksDB Backends

	Heap	RocksDB
Stores:	Java instances	Serialized bytes
Serializes:	On savepoint	On state update



Creating a savepoint



Heap Backend

Registered TS

Create TSS

Serialize TSS

Serialize instance

RocksDB Backend

Registered TS

Create TSS

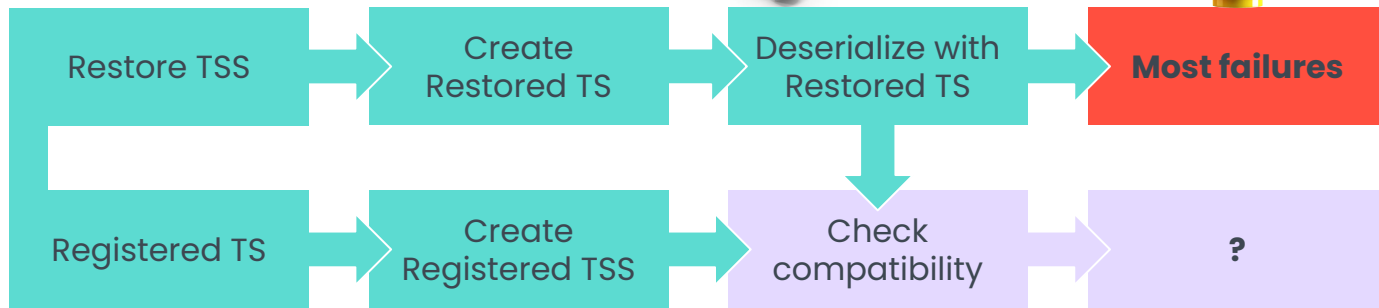
Serialize TSS

Copy bytes

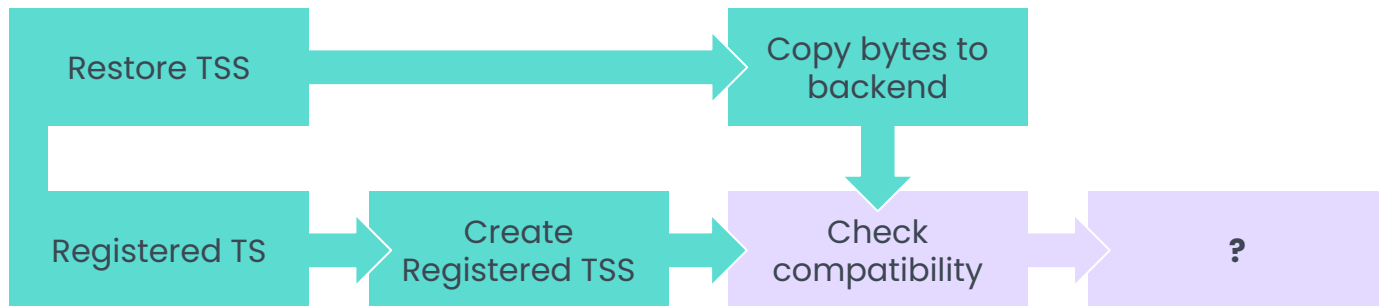
Restoring a savepoint



Heap Backend



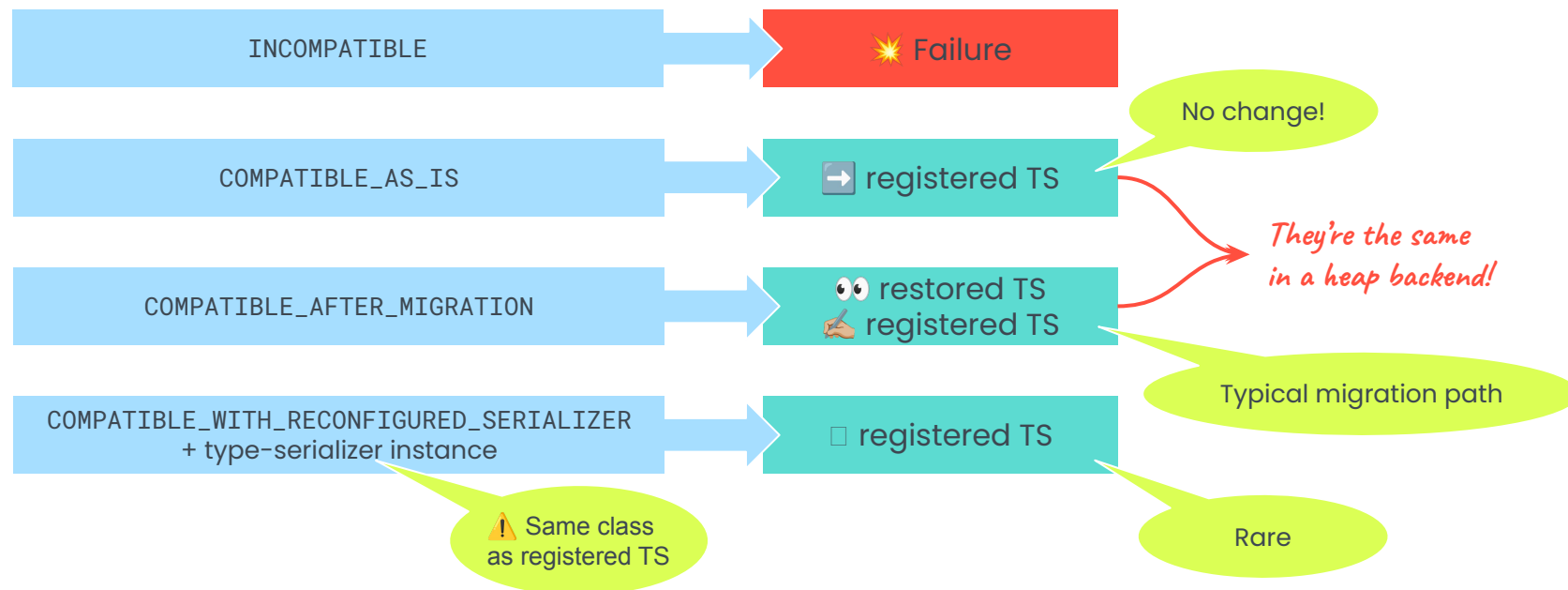
RocksDB Backend



registered

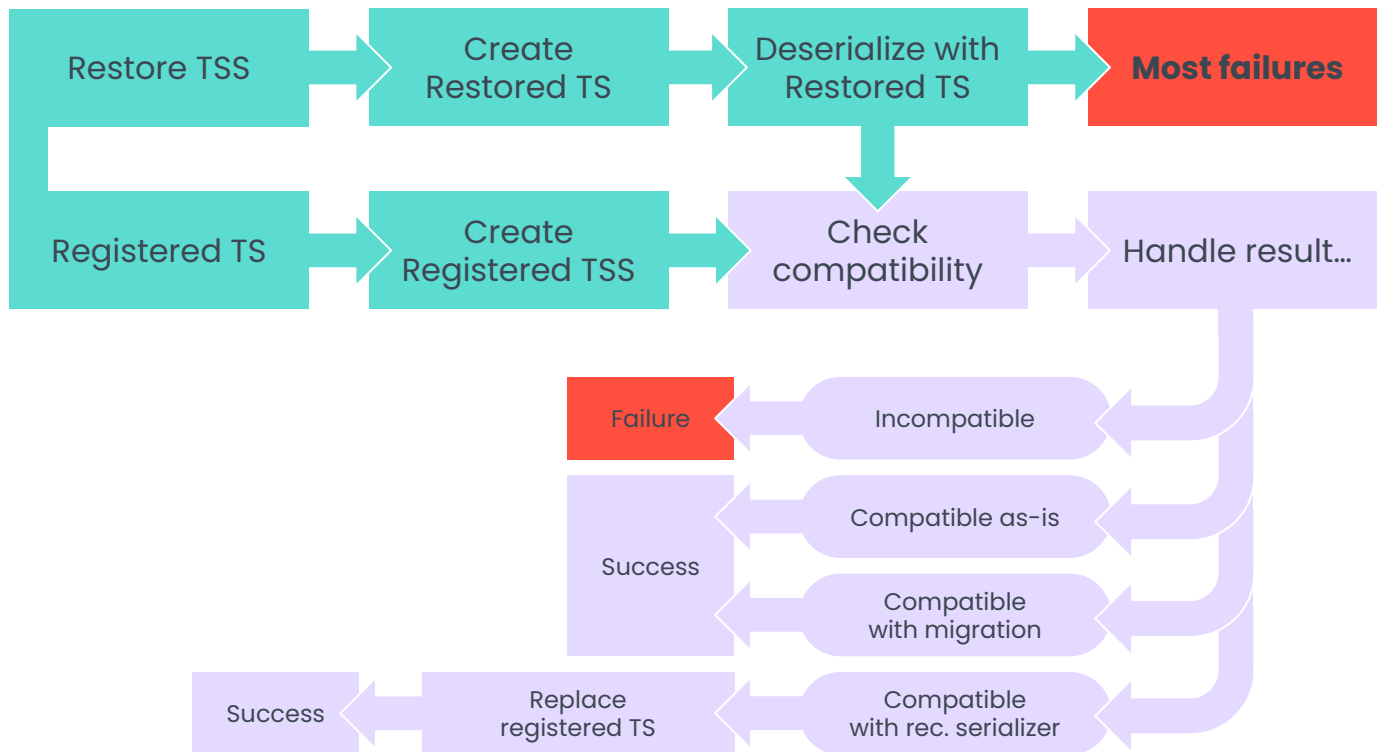
Checking compatibility

```
registeredSnap.resolveSchemaCompatibility(recoveredSnap);
```



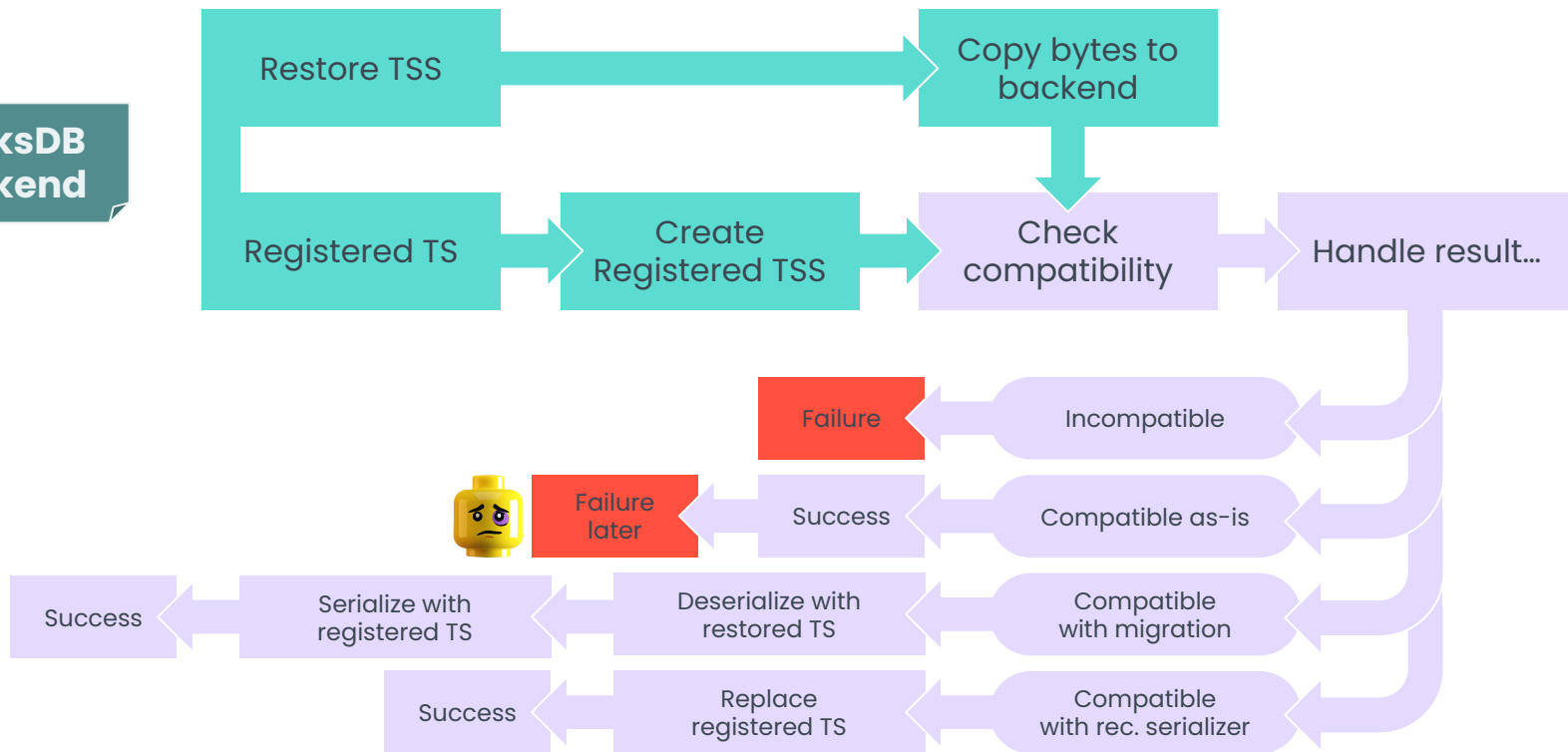
Restoring a savepoint

Heap
Backend



Restoring a savepoint

**RocksDB
Backend**



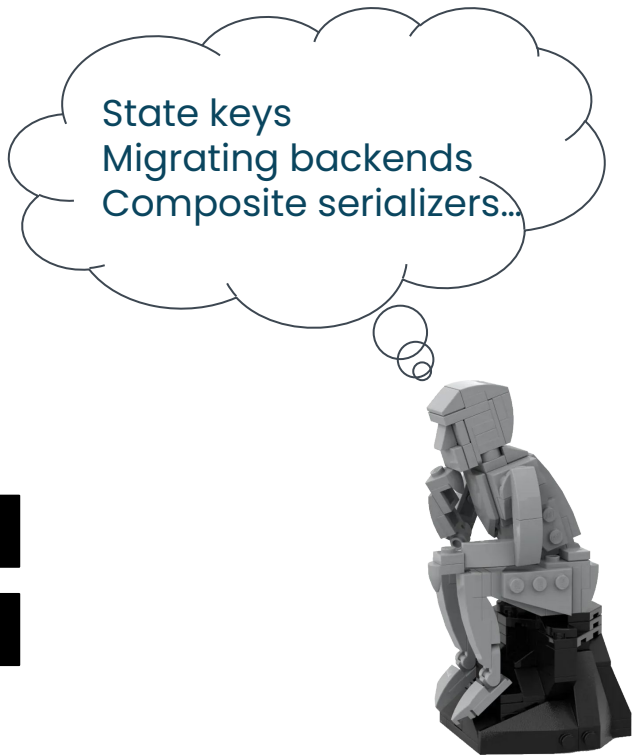
Learn more



State Schema Evolution
Custom Serialization for Managed State

```
HeapKeyedStateBackend.tryRegisterStateTable(...)
```

```
RocksDBKeyedStateBackend.updateRestoredStateMetaInfo(...)
```



Evolution use cases

POJO serializer: supported evolutions



Remove fields



Add fields



Reorder fields



From POJO to custom serializer

```
class Count {  
    private String ip;  
    private long totalCount;  
    private int errorCount;  
    ... // no-arg constructor + getters + setters  
}
```

Default PojoSerializer

```
COUNT_DESCRIPTOR = new ValueStateDescriptor<Count>( "count", Count.class);
```

Custom Serializer

```
COUNT_DESCRIPTOR = new ValueStateDescriptor<Count>( "count", new CountSerializer());
```

Change the type of a field

Restored savepoint

"CountSerializerSnapshot"

version = 1

Some Count fields

int errorCount

CountSerializer

```
Count deserialize(Count count, DataInputView source) {  
    ...  
    if (this.version == V1) {  
        count.setErrorCount(source.readInt());  
    } else {  
        count.setErrorCount(source.readLong());  
    }  
    return count;  
}
```

Read as a int

CountSerializerSnapshot

```
TSSchemaCompatibility<Count>  
resolveSchemaCompatibility(TSS<Count> restored) {  
    if (this.version == V2 &&  
        restored.version == V1) {  
        return compatibleAfterMigration();  
    } else ...  
}
```

Next savepoint

"CountSerializerSnapshot"

version = 2

Some Count fields

long errorCount

*Both must be true to
trigger the migration path*

Rename a state class

Restored savepoint

"CountSerializerSnapshot"

version = 2

Count fields

CountSerializerSnapshot

```
TypeSerializer<Count> restoreSerializer() {  
    return new StatsSerializer(version);  
}
```

StatsSerializerSnapshot

```
TypeSerializerSchemaCompatibility<Stats>  
    resolveSchemaCompatibility(TSS<Stats> restored) {  
  
    if (restored instanceof StatsSerializerSnapshot ||  
        restored instanceof CountSerializerSnapshot) {  
        return compatibleAsIs();  
    }  
    ...  
}
```

Next savepoint

"StatsSerializerSnapshot"

version = 2

Stats fields

Accept old and new name

Work-in-progress: evolution annotations

```
@version(3)
@deletedFields(since = 2, "timestamp")
@renamed(since = 2, "Count")
case class Stats(
  ip: String,
  @renamed(since = 3, "total") totalCount: Long,
  @transformed(since = 2, mapIntToLong) errorCount: Long,
  @added(since = 3) redirectCount: Long = 0
)

def mapIntToLong(value: Int): Long = value.toLong
```

Coming soon... in flink-scala-api



Unit testing

State validation

Run test (1st time)

Create savepoint in src/test/resources

Committed to Git

Re-run test

Compare with existing savepoint



No diff



Compatible changes



Incompatible

*Test is failed (yellow)
Might be just content!
Check if expected*

Test in error (red)

Using Flink's
test harnesses
(AbstractStreamOperator
TestHarness)

With a wrapper because 🤔

Coming soon: state validation harness fixture

```
trait Fixture extends FixtureTestSuite {  
  
  type FixtureParam <: Harness[_]_  
  
  def makeHarness(savepointFilePath: Option[String] = None): FixtureParam  
  
  override def withFixture(test: OneArgTest): Outcome = {  
    val harness: FixtureParam = makeHarness()  
    try {  
      val outcome = withFixture(test.toNoArgTest(harness))  
      if (outcome.isSucceeded) {  
        val validation = validate(harness, test.name, makeHarness(Some(_)))  
        handleValidation(validation)  
        outcome  
      } else outcome  
    } finally harness.close()  
  }  
}
```

Implementation in ScalaTest

```
class ProcessFunctionHarnessTest extends TestSuite with HarnessStateValidationFixture
{
  override def makeHarness(savepointFilePath: Option[String] = None): Harness =
    ProcessFunctionHarness(new MyFunction, savepointFilePath)

  "Process message and read" in { harness =>
    harness.process(42)
    harness.process(100)
    harness.getOutput should be(List(42, 100))
  }
}
```

Coming soon... in flink-scala-api

Thank you! Any Questions?

<http://datadome.co>

